Project Manager Maneuver Ammunition Systems

Product Lead Special Ammunition and Weapon Systems

GENERAL SPECIFICATION FOR SPECIAL AMMUNITION and WEAPON SYSTEMS

10 February 2026

1. SCOPE

1.1 Scope. This general specification describes the requirements and verification procedures for Special Ammunition and Weapon Systems excluding Small Caliber ammunition, Mortar ammunition, Mortar Weapon Systems, and Rocket Propelled & Spin Stabilized Grenade ammunition (see family specifications for these items).

2. APPLICABLE DOCUMENTS

2.1 General. In addition to documents listed in this Section, there are documents specified in Section 5 of this specification. All documents listed in this specification should use the latest revision available at the time of award.

2.2 Government documents. Not Applicable.

2.3 Non-Government publications.

a) United Nations Recommendations on the Transport of Dangerous Goods Model Regulations, 24th Edition, 2025.

b) International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air, 2025-2026 Edition.

c) International Maritime Organization (IMO) International Maritime Dangerous Goods Code (IMDG) 2024 Edition, Amendment 42-24.

2.4 Acquisition requirements. Acquisition documentation (base award and/or Request for Proposal (RFP)) must specify the following:

a. Title and date of this specification.
b. Requirements for certificates of conformance for each lot or shipment of product.
c. Requirements for age of ammunition.
d. Requirements and provisions for submission of data as required.
e. Requirements for acceptance criteria if different than those stated in specification.
f. Requirements for reduction or elimination of verification procedures.
g. Requirements and provisions for contractor and Government verification.
h. Requirements and provisions for packaging of ammunition.
i. Requirements and provisions for transportation of ammunition.

2.5 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Requirements specified in the contract or delivery order precede the text of this document. Nothing in this document or the contract, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Technical requirements. Ammunition and non-energetic items shall meet the requirements and verifications of this specification for the ammunition/items listed in ANNEX A. All ammunition or items shall be accompanied with technical data (see APPENDIX) that meets or exceeds the requirements of this specification.

3.1.1 Compliance to technical requirements and storage history records. A Technical Data Package (TDP) as well as lot history folders shall be provided to the U.S. Government (USG) to ensure ammunition/items are in compliance with this specification. Additionally, documented storage history of the ammunition/items shall be provided.

3.2 Conformance Testing and Inspection. A random sample of the ammunition/items from each lot presented shall be selected by a USG Representative and subjected to conformance testing and inspection in accordance with TABLE I (to include retest rounds, if required).

3.2.1 Stock ammunition. All ammunition/items must be less than 5 years old and meet the verification procedures in TABLE I of this specification. Any lot that has not been adequately tested to comply with all requirements of TABLE I shall be subjected to conformance testing and inspection in accordance with this specification.

3.3. Serviceability. Ammunition/items shall be serviceable and issuable, in good condition and without visible signs of degradation. All parts and assemblies shall be fabricated and finished in a thorough matter and shall be free of burrs, chips, dirt, grease, rust, corrosion, and other foreign material that adversely affect fit or function. Coatings and protective finishes shall be uniform and shall provide complete coverage of designated surfaces. The painting and marking shall be restored by the assembling facility on any components that have been marred or obliterated.

3.3.1 Identification of defects. Visual defects shall be identified within the TDP and include a defined method for acceptance/rejection of the ammunition/items. Visual inspections listed in ANNEX B shall be the minimum requirement.

3.3.2 Weight zones. A table outlining weight ranges by zone is required in the TDP if there are multiple projectile weight zones.

3.4 Weapon interface and interoperability. The ammunition/items shall be functionally compatible with the weapons identified in ANNEX A or as otherwise specified in the contract or delivery order.

3.5 Operating requirements. The ammunition/items shall provide their functional, operational, and performance capabilities when fired regardless of weather-related conditions (example: rain, snow, mud, wind, fog). If testing occurs in inclement weather, the Contractor shall implement all necessary safety precautions.

3.5.1 Operating temperatures. The ammunition/items shall demonstrate safety and reliability throughout exposure to ambient and extreme temperatures. All cartridges are required to reliably perform within a temperature range of -30°C to +50°C.

3.5.2 Muzzle velocity. The average velocity of the ammunition obtained at ambient temperature shall meet the requirement listed in the TDP. Each lot shall meet the standard deviation requirements. The value of the deviation if not provided in Request for Proposal (RFP) will be determined during TDP development.

3.5.3 Chamber pressure. The individual maximum and average pressure of the ammunition obtained at ambient and hot shall meet the requirements specified in the TDP.

3.5.4 Maximum effective range. Each lot of ammunition shall be tested at ambient to determine maximum range as listed in the TDP.

3.5.5 Dispersion. The ammunition shall demonstrate dispersion characteristics at ambient as specified in the TDP.

3.5.6 Function and casualty. The ammunition/items shall function without casualty or cause damage to the weapon, and shall exhibit a ballistic performance free of misfires, duds, premature bursts, short rounds, and metal parts separation. The ammunition/items shall cause no hazard or unsafe condition to the gunner during storage, transportation, or use.

3.5.7 Performance. The ammunition/items shall function according to its specific purpose and as defined in the TDP. Specific performance requirements may include but are not limited to fragmentation, airburst, bounding, proximity, heat seeking, homing, illumination, penetration, smoke and flare.

3.6 Fuze identification and description. When applicable, the fuze shall be identified within each TDP with a drawing/figure of the internal fuze design with components categorized to include the type and weight of energetic material(s) and a functional description of the fuze that includes, at a minimum, a detailed description of the fuze arming sequence, operating temperatures, in-process tests (example: jolt/jumble, sensitivity, safe arming/non-arming etc.), and no arm and all arm distances. Test records shall be available to the USG for verification. Fuze shall be set to safest setting, if applicable.

3.7 Environmental Requirements.

3.7.1 Sequential rough handling. Ammunition/items shall be capable of withstanding the rigors of the sequential rough handling and transportation throughout extreme temperature ranges and meet all performance and safety requirements.

3.7.2 Shelf life. The ammunition/items shall meet all requirements of this specification within its operating temperature range after a minimum of ten (10) years of storage. All cartridges are required to reliably be stored within a temperature range of -30°C to +50°C at a minimum of ten (10) years.

4. VERIFICATION

4.1 Technical verification. The ammunition/items shall be subjected to verification of all requirements cited in Section 3 in this specification, in accordance with TABLE I. Noncompliance with any requirements shall be cause to withhold acceptance of the lot or batch in which the noncompliance was found.

TABLE I. Requirements/Verification Cross-Reference Matrix

Section 3 Requirements	Requirement	Section 4 Verification Procedures	Sample Size
3.1	Technical requirements	4.1	-
3.3	Serviceability	4.3	Note 1
3.3.1	Identification of defects	4.3	ANNEX B
3.4	Weapon interface and interoperability	4.4	ANNEX A
3.5.1	Operating temperatures	4.5.1	ANNEX A
3.5.2	Muzzle velocity	4.5.2	ANNEX A, Note 2
3.5.3	Chamber Pressure	4.5.3	ANNEX A, Note 2
3.5.4	Maximum effective range	4.5.4	ANNEX A, Note 2
3.5.5	Dispersion	4.5.5	ANNEX A, Note 2
3.5.6	Function and casualty	4.5.6	ANNEX A
3.5.7	Performance	4.5.7	ANNEX A
5.1	Packaging	-	ANNEX B
5.3	Packing	-	ANNEX B
5.5	Marking	-	ANNEX B

Note 1: To be performed using a defined sampling procedure for inspection determined by the lot sizes.

Note 2: Not applicable for non-energetic items such as links.

4.1.1 Lot formation. The ammunition/items shall be assembled into identifiable lots, sublots, or batches, or in such other manner contingent on approval by USG. Each lot or batch shall, as far as practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time. Lot quantities shall be determined by the total quantity for shipment and include the additional rounds from the same lot to be consumed during lot acceptance testing along with contingency rounds for potential retesting. The lots or batches shall be identified by the contractor and shall be kept intact in adequate and suitable storage space. All ammunition in a given lot must be manufactured prior to lot acceptance testing.

4.1.1.1. Lot Size. All lot sizes are to remain consistent throughout an awarded delivery order. Once temperature testing is conducted, all skip lot sizes shall be within 20% of that lot's size. If a lot size larger than 20% is proposed, temperature testing has to be conducted on that lot. If a lot fails for any reason, the next two lots presented for additional testing have to be within 20% of the failed lot size.

4.1.1.2 Component Lots. If multiple lots of a component are present, a minimum of 10 rounds shall be fired for each lot.

4.1.2 Retest criteria. A ballistic retest is authorized upon initial failure of ammunition/items to meet test requirements as long as the failure is not considered critical. Retest procedures must be clearly defined in the TDP and results must be combined with data from the first test unless a failure analysis confirms that the failure was caused by something other than the ammunition/items being tested (example: weapon failure or faulty test set up).

4.2 Verification of compliance. Evidence shall include, but is not limited to, the following;

a. Identification of technical requirements to which the ammunition/items is/was produced.
b. Manufacturer, date of manufacture, and original acceptance documentation.
c. Initial acceptance reports, including type, lot or batch identification, quantity and method of acceptance (example: sample size, verification method, acceptance criteria, results…).
d. Surveillance reports, including lot or batch identification, quantity, and method of surveillance (example: sample size, verification method, criteria for action to be taken on lot or batch, results).
e. Storage history, including duration, and storage condition (example: controlled, uncontrolled).

4.3 Serviceability and identification of defects. A random sample shall be selected from each lot of ammunition/items for inspection. The ammunition/items shall be visually inspected in accordance with ANNEX B of this specification. Reinspection of a lot that fails for non-ballistic reasons is authorized as long as the criteria for reinspection is clearly outlined in the TDP per ANNEX B.

4.3.1 Weight zone verification. For items with multiple projectile weight zones, 24 samples must be weighed and fall within the allotted ranges in the weight zone table provided in the TDP.

4.4 Weapon interface and interoperability. A random sample shall be selected from each lot of ammunition/items to be delivered and inserted into a weapon chamber or chamber gage that conforms to the identified weapon system to check for profile and alignment. Inability to interface properly with the weapon system shall be considered a failure to show compliance and constitute a single reliability failure.

4.5 Operating verification. The same cartridge of ammunition can be used for multiple tests provided the test equipment is capable of capturing all necessary test data. For example: Muzzle Velocity, Maximum Effective Range, and Function & Causality testing can be done at the same time using the same cartridges. Called warmers and spotters are allowed and encouraged. “Called” means that before the cartridge is loaded into the weapon, the USG representative must be informed that the next cartridge to be fired is a warmer or spotter. Should a cartridge not be "Called" it shall be counted toward whichever test is being conducted, potentially resulting in a LAT failure. Critical safety failures apply to all cartridges fired during LAT, regardless of which test is being performed. They shall result in the immediate rejection of the lot and there shall be no retest. Critical safety failures are listed in Annex A and are specific to each ammunition/item family.

4.5.1 Operating temperatures. Ammunition shall operate reliably and safely when conditioned and fired throughout a reasonable extreme operating temperature range. Ammunition shall consistently perform within a temperature range of -30°C to +50°C, ensuring reliable function under the specified environmental conditions. With the exception of hand grenades, temperature testing may be conducted on a skip-lot basis (see APPENDIX) during LAT. A single critical safety failure will result in lot rejection and there shall be no retest. Functional testing parameters must be defined in the TDP and meet manufacturer requirements. The entire cartridge, including all components necessary to fire and function (example: fuze, warhead, propellant, etc.) shall be temperature conditioned based on the maximum cartridge body diameter as described below. Additionally, the cartridges shall be fired within the following times after being removed from the conditioning chamber or thermally insulated bag.

a. Ammunition less than 85mm in diameter shall be conditioned for not less than six (6) hours and fired within five (5) minutes of removal from conditioning box or thermally insulated bag.
b. Ammunition between 85mm and 120mm in diameter shall be conditioned for not less than nine (9) hours and fired within six (6) minutes of removal from conditioning box or thermally insulated bag.
c. Ammunition greater than 120mm in diameter shall be conditioned for not less than twelve (12) hours and fired within eight (8) minutes of removal from conditioning box or thermally insulated bag.

4.5.1.1 Projectile weight and charge operational test criteria: If there are multiple projectile weight zones, the lowest weight zone available shall be conditioned and tested at the lowest operating temperature and the highest weight zone available will be conditioned and tested at the highest operating temperature. The highest weight zone will also be utilized for ambient testing. In addition, the minimum propellant charge zone shall be used at the lowest operating temperature, and the maximum propellant charge zone shall be used at the highest operating temperature.

4.5.2 Muzzle velocity. Velocity tests shall be conducted in accordance with the manufacturer's test procedures to verify that a sample representative of the lot meets the minimum velocity requirements specified in the TDP (10 cartridges minimum). If the lot fails to meet the Velocity requirements a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes. Each lot shall meet the standard deviation requirements specified in the RFP or TDP.

4.5.3 Chamber pressure. Pressure tests shall be conducted in accordance with the manufacturer's test procedures to verify samples from the lot do not exceed the average or individual pressure requirements listed in the TDP at ambient (10 cartridges minimum) and hot (10 cartridges minimum). Each lot of propellant charges shall meet the individual maximum and average pressure requirements specified in the RFP or TDP. Pressure testing shall be conducted during the LAT for all Artillery, Medium Cal, and Tank ammunition. Pressure testing can be done in process for Rocket and ATGM ammunition. Pressure testing is not required for hand grenades, pyro, links, linked belts, or linking machines. If the lot fails to meet the Pressure requirements a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes. Any single round that exceeds the maximum individual pressure requirement will result in lot rejection; there shall be no retest.

4.5.4 Maximum Effective Range. Maximum effective range shall be verified during lot acceptance testing to ensure compliance with the TDP (does not apply to medium caliber ammunition).

4.5.5 Dispersion. Dispersion tests shall be conducted to verify that each lot of ammunition meets the specified requirements in the TDP (10 cartridges minimum). If the lot fails to meet the Dispersion requirements or a round misses the target, a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum or 19 cartridges if one misses the target), shall be used to determine if the lot passes. If two (2) or more cartridges miss the target (cumulative) the lot fails.

4.5.6 Function and casualty. A random sample shall be selected from the ammunition/items lot in accordance with ANNEX A to be delivered and functioned from the identified weapon system to check for function, casualty and metal parts security. The ammunition/items shall function across the entire operating temperature range (extreme hot, cold, ambient) without casualty or damage to the weapon, and shall exhibit a ballistic performance free of misfires, duds, premature bursts, short rounds, and metal parts separation. A short round is defined as a cartridge that has half of the muzzle velocity of the others within its group and/or impacts the ground at a % distance less of the range average or its respective temperature/charge/group, excluding the cartridge(s) in question. Short rounds are critical safety failures for Artillery cartridges if they impact at 80% or less. Short rounds are function failures for all other types of ammunition if they impact at 50% or less. The ammunition shall cause no hazard or unsafe condition to the gunner during storage, transportation, or use. If a cartridge has a multi-option fuze, all options shall be tested during Lot Acceptance Testing to properly demonstrate the fullest extent of the munitions capability unless specified otherwise in Annex A or by the USG during solicitation. The different fuze options test quantity will be distributed amongst the total Function and Casualty requirement. The ammunition shall cause no hazard or unsafe conditions to the user during storage, transportation or use. Critical safety failures are listed in Annex A of this specification and are specific to each family of ammunition.

4.5.7 Performance. Proper functioning for each ammunition/items type is defined below. Any round which does not function properly according to the definitions below and the requirements in the TDP are considered a function failure:

a. Fragmentation: The ammunition, with a point detonate fuze, shall produce a high-order detonation of the warhead upon initial impact with the target (terrain or man-made structure) as specified in the TDP. If fuze contains a delay the ammunition shall detonate after impact with target at a specific time or distance as specified in the TDP.
b. Illumination: The ammunition shall produce parachute suspended light at specific intensity, duration, wavelength, and height as specified by the manufacturer in the TDP.
c. Smoke: The ammunition shall produce a volume of smoke upon impact with the target as specified by the manufacturer in the TDP.
d. Bounding: The ammunition shall bound/jump after initial impacting with the ground and produce a high-order detonation of the warhead above ground between a low/high height as specified by the manufacturer in the TDP.
e. Airburst: The ammunition shall produce a high-order detonation of the warhead above ground between a min/max distance as specified by the manufacturer in the TDP.
f. Practice (Inert): The ammunition shall not contain any explosives or energetics other than propellant.
g. Practice (Live): The ammunition may contain energetics, other than propellant, to signal location of impact.
h. Proximity: The ammunition shall produce a high-order detonation of the warhead within a min/max distance of target as specified by the manufacturer in the TDP.
i. High-Explosive Anti-Tank (HEAT): The ammunition shall produce a high-order detonation resulting in jet capable of penetrating armor at a depth as specified in the TDP.

j. Heat Seeking: The ammunition shall detect and navigate itself to a target heat source and function as specified by the manufacturer in the TDP.
k. Homing: The ammunition shall navigate itself to a target and function as specified by the manufacturer in the TDP. (example: GPS, Radar, etc.)
l. Penetration: The ammunition shall meet the armor penetration characteristics as specified in the TDP.
m. Flare: The ammunition shall deploy from the launcher and produce its intended effect for a minimum amount of time as specified in the TDP.
n. Links and Linked Belts: Link and belt requirements shall include applicable testing for cartridge extraction forces, fanning, folding, twisting, loading, extension, failure free feeding in respective weapon, etc. as specified in the TDP.
o. Linking Machines: Linking machines shall include all applicable testing to verify they fully seat the cartridge into the belt without damage to the cartridge, belt or linking machine as well as smooth failure free functioning without binding, interference or loss of motion as specified in the TDP.

5. PACKAGING

5.1 Preservation, packaging, packing, unitization, and marking shall provide protection for multiple handling, redistribution, and shipment by any transportation mode and meet or exceed the following requirements.

5.1.1 Packaging containers for hazardous materials, ammunition and explosives shall meet or exceed the requirements found in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.1.2 Cleanliness – Items and packaging shall be free of dirt and other contaminants which would contribute to the deterioration of the item, or which would require cleaning by the customer prior to use. Coatings and preservatives applied to the item for protection are not considered contaminants.

5.1.3 Preservation – Items susceptible to corrosion or deterioration shall be provided protection against external environmental effects.

5.1.4 Cushioning – Items requiring protection from physical and mechanical damage (example: fragile, sensitive, critical material) or which could cause physical damage to other items, shall be protected by wrapping, cushioning, pack compartmentalization, or other means to mitigate shock and vibration and prevent damage during handling and shipment.

5.1.5 Wood Packing Material (WPM) – All natural wood packing material shall be constructed from heat treated lumber and certified by an accredited recognized agency in accordance with International Standards for Phytosanitary Measures (ISPM-15) regardless of origin or destination. All natural wood packing materials must include certification markings that are readily visible to inspectors and on at least two sides of every container when palletized.

5.1.6 Firing Tables – A copy of the manufacturer’s firing tables for ammunition shall be included with the TDP.

5.2. Unit Package

5.2.1 Unit package shall be so designed and constructed that it will contain the contents with no damage to the item(s), and with minimal damage to the unit pack during shipment and storage in the shipping container and will allow subsequent handling. The outermost component of the unit package shall be a container such as a sealed bag, carton, or box.

5.3. Packing

5.3.1 Unit packages must be packed in shipping containers. All shipping containers shall be the most cost effective and shall be of the minimum cube to contain and protect the items.

5.3.2 Shipping Containers – The shipping container (including any necessary blocking, bracing, cushioning, or waterproofing) shall comply with the regulations of the carrier used and shall provide safe delivery to the destination at the lowest tariff cost. The shipping container shall be capable of multiple handling, stacking at least ten feet high, and storage under favorable conditions and meet the requirements of the “United Nations Recommendations on the Transport of Dangerous Goods”.

5.4. Unitization

5.4.1 Shipments of identical items going to the same destination shall be palletized if they have a total cubic displacement of 20 cubic feet or more unless skids or other forklift handling features are included on the containers. Pallet loads must be stable, and to the greatest extent possible, provide a level top for ease of stacking. The weight capacity of the pallet must be adequate for the load. A pallet load shall not exceed 4,000 pounds and should not exceed 52 inches in length or width, or 54 inches in height. The load shall be contained in a manner that will permit safe handling during shipment and storage.

5.4.2 Banding – Metal banding shall be used to secure the load. Straps shall be applied to each column or layer of boxes. Tie down straps shall be applied to each column of boxes at 90 degrees to the load straps. Edge protectors shall be used when securing fiberboard boxes.

5.5. Marking

5.5.1 Packaging marking shall be visible, clear, and remain legible during normal life cycle handling.

5.5.2 All unit packages, intermediate packs, exterior shipping containers, and as applicable, unitized loads shall be marked with item description, quantity, lot number, or serial number. The outer shipping container and unitized load shall indicate load weight, UN dangerous goods proper shipping name, and location of UN number and UN POP marking.

5.5.3 Each package shipping container shall show the United Nations packaging symbol and applicable codes in accordance with the construction requirements and testing of packaging as expressed in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations".

5.6. Additional Requirements for Hazardous Materials

5.6.1 The shipment shall be prepared in accordance with the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and other applicable regulations effective at the time of shipment in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.6.2 Packaging and marking for hazardous material shall comply with the requirements for the mode of transport and the applicable performance packaging contained in the following documents:

a. International air transport: International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air.
b. International vessel transport: International Maritime Organization (IMO) International Maritime Dangerous Goods Code (IMDG).

6. REPORTING

6.1 Lot Acceptance Test Reports (LATR), at a minimum, must include a summary of findings for all visual and ballistic testing, lot history folders (in-process testing per TDP requirements, ammo data card, and all certifications or certificates), all measured data (including weights), pressure and velocity for each individual shot, range for all test shots, deflection for all test shots scored for dispersion, identify conditioning temperature used, calculated average pressure and velocity, spread for minimum and maximum velocity, and calculated dispersion. The LATR must also include the ammo data card for all lots tested.

6.1.1 Ammo Data Cards are required to be provided as part of the LATR and must include lot information for all energetics and major components used in the production of each lot.

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Annex A: Weapon Interface and Sample Size Requirements
Annex B: Visual Inspection Requirements
APPENDIX - Definitions

ANNEX A - Weapon Interface and Sample Size Requirements

Medium Caliber					
Item[1,8,9]	Weapon Interface[11]	Function and Casualty/Safety Order size greater than 1000 cartridges Sample Size[2]-Accept-Reject & Retest (cumulative)			Function and Casualty/Safety Order size 100-1000 cartridges[10] Sample Size[2]-Accept-Reject & Retest (cumulative)
		Ambient	Hot	Cold	Ambient Only
14.5 x 114mm (All Types)[3,4,5,6,7]	KPV Machine Gun	32-0-1 Retest 60-1-2 (Cumulative)	32-1-2 Retest 60-2-3 (Cumulative)	32-1-2 Retest 60-2-3 (Cumulative)	32-0-1 Retest 60-1-2 (Cumulative)
23 x 115mm (All Types)[3,4,5,6,7]	GSh-23				
23 x 152mm (All Types)[3,4,5,6,7]	ZU-23				
30 x 165mm (All Types)[3,4,5,6,7]	2A42 or Gsh-30				
Notes on next page					

Note 1: Any medium caliber ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show function and casualty at ambient temperature. 32 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: Not less than 80% of the Tracer cartridges (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's TDP. All tracers within the lot shall be of the same color.

Note 4: Not less than 80% of the Armor piercing cartridges (10 shot minimum) shall have full penetration of armor with a thickness as per the manufacturer's TDP.

Note 5: Not less than 80% of the Incendiary cartridges (10 shot minimum) shall, upon impact, develop an incandescent flash sufficiently hot, intense and sustained to ignite explosive vapors or initiate combustion in readily ignitable materials.

Note 6: Not less than 80% of High Explosive cartridges (10 shot minimum) shall completely detonate upon initial impact with their target. Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test.

Note 7: If the lot fails to meet the Tracer, Incendiary, Armor Piercing, and/or HE requirements, a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes.

Note 8: A function failure shall be defined as:

a) A cartridge that will not load into the weapon.
b) A misfire - failure of a projectile to discharge as intended.
c) A short round: A cartridge that has half of the muzzle velocity of the others within its group and/or impacts the ground at a 50% less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
d) A cartridge that does not completely detonate upon initial impact with their target (HE and Practice cartridges only). Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test.
f) A cartridge that does not perform its primary function

Note 9: Critical safety failures are defined as:

a) Detonation, Deflagration, or function within the weapon (all cartridge types).
b) Detonation of an HE or HEAT cartridge before impact with the ground or target.
c) An early function of Practice (Live), Smoke, Incendiary, Airburst or Illumination cartridge which may expose the operator to smoke, fragmentation or fire.
d) Metal parts separation.
e) A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

Note 10: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 11: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

Artillery Ammunition					
Item[1,4,6,7]	**Weapon Interface[10]**	**Function and Casualty/Safety Order size greater than 250 cartridges Sample Size[2]-Accept-Reject & Retest (cumulative)**			**Function and Casualty/Safety Order size 100-250 cartridges[3] Sample Size[2]-Accept-Reject & Retest (cumulative)**
		Ambient	Hot	Cold	Ambient Only
122mm HE Frag / HEAT[5] 122mm Airburst 122mm Smoke 122mm Illumination	D-30 Towed Howitzer	15-0-1 Retest 29-1-2 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)
152mm Frag-HE[5]	D-20 Towed Howitzer				
155mm HE	M114, M777 and M109 Howitzers				
105mm HE	M1, M2, M2A1, M101, M102 and M119				
122mm Artillery Rockets[5]	BM-21 Grad				
Notes on next page					

Note 1: Any artillery ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 15 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 4: The average corrected range of all cartridges when discharged during ambient testing shall be within 95% to 105% of their respective firing tables.

Note 5: For these cartridge types only, if they contain a multi-option fuze where the only options are Point Detonate and Delay, they shall be set to Point Detonate during LAT.

Note 6: A function failure shall be defined as:

a) A cartridge that will not load into the weapon.
b) A misfire - failure of a projectile to discharge as intended.
c) A cartridge that does not produce smoke (Smoke cartridges only).
d) A cartridge that does not have a parachute suspended burn time as identified in the TDP. (Illumination cartridges only).
e) A cartridge that does not completely detonate upon initial impact with their target (HE and Practice cartridges only). Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test.
f) A cartridge that does not perform its primary function

Note 7: Critical safety failures are defined as:

a) Detonation, Deflagration, or function within the weapon (all cartridge types).
b) Detonation of an HE or HEAT cartridge before impact with the ground or target.
c) An early function of Practice (Live), Smoke, Airburst or Illumination cartridge which may expose the operator to smoke, fragmentation or fire.
d) Short rounds; a cartridge impacts the ground at a range 80% or less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question (HE, HEAT & Smoke only). An Airburst cartridge that detonates within the weapon or functions at a range of 80% or less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
e) Metal parts separation.
f) A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

Note 8: For 155mm HE, simulation pressure testing is required at ambient if the supplier uses a weapon system with a chamber volume larger 18.7 liters. Five additional test rounds must use charges either modified or of an alternate configuration to simulate the pressure that would be obtained in the M777 or M109 system. Pressure will be adjusted based on weapon system used during the LAT. The following equation will be utilized to calculate the minimum pressure of the 5 pressure simulation test rounds. Achieved Average Pressure (Ambient) x (A/18.7) where A is the chamber volume of the alternate 155mm weapon system. At 19.9 liters chamber volume, minimum pressure is 28,082 psi and at 23 liters, minimum pressure is 31,425 psi. All 5 rounds must function properly with no safety failures. If there is one function failure

Note 9: For 105mm HE, simulation pressure testing is required at ambient if the supplier uses a weapon system with a chamber volume larger than 153 cubic inches. Five additional test rounds must use charges either modified or of an alternate configuration to simulate the pressure that would be obtained in the M777 or M109 system. Pressure will be adjusted based on weapon system used during the LAT. Each of the 5 test rounds shall have a test chamber pressure of not less than 42,700 psi. All 5 rounds must function properly with no safety failures.

Note 10: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

Note 11: Pressure simulation testing can be required for other artillery systems based on chamber volume. Specific requirements will be provided in the RFP.

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

<table>
<tr><th colspan="6">Aircraft Ammunition – Aviation Rockets</th></tr>
<tr><th rowspan="2">Item[1,4,6]</th><th rowspan="2">Weapon Interface[7]</th><th colspan="3">Function and Casualty/Safety
Order size greater than 250 cartridges
Sample Size[2]-Accept-Reject & Retest (cumulative)</th><th>Function and Casualty/Safety
Order size 100-250 cartridges[3]
Sample Size[2]-Accept-Reject & Retest (cumulative)</th></tr>
<tr><th>Ambient</th><th>Hot</th><th>Cold</th><th>Ambient Only</th></tr>
<tr><td>57mm S-5KO
57mm S-5KP
57mm S-5MO</td><td rowspan="2">UB-16-57UM/
UB-32
Rocket Pod</td><td>15-0-1
Retest 29-1-2
(Cumulative)</td><td rowspan="3">15-1-2
Retest 29-2-3
(Cumulative)</td><td rowspan="3">15-1-2
Retest 29-2-3
(Cumulative)</td><td>15-0-1
Retest 29-1-2
(Cumulative)</td></tr>
<tr><td>57mm S-5 Practice</td><td rowspan="2">15-1-2
Retest 29-2-3
(Cumulative)
Graze Function Only[5]</td><td rowspan="2">15-1-2
Retest 29-2-3
(Cumulative)
Graze Function Only[5]</td></tr>
<tr><td>80mm S-8
80mm S-8KO
80mm S-8D</td><td>B-8M1/B8V20
Rocket Pod</td></tr>
<tr><td colspan="6">Notes on next page</td></tr>
</table>

Note 1: Any aviation ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 15 rounds shall be tested for function and safety temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 4: A function failure shall be defined as:

a) A cartridge that will not load into the weapon.
b) A misfire - failure of a projectile to discharge as intended.
c) A short round: A cartridge that has half of the muzzle velocity of the others within its group and/or impacts the ground at a 50% less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
d) A cartridge that does not completely detonate upon initial impact with their target (HE and Practice cartridges only). Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test.
e) A cartridge that does not perform its primary function.

Note 5: The Function and Casualty Accept/Reject criteria for low angle graze function is 15-1-2, Retest 29-2-3 (cumulative). A graze function failure is defined as failure of fuze detonation due to a low angle of impact of less than 30 degrees. A photo of the test setup and target area must be included in the TDP to verify the low angle of impact and must be approved by the USG during the TDP approval process. This test is conducted in conjunction with normal function and casualty using the same cartridges.

Note 6: Critical safety failures are defined as:

a) Detonation, Deflagration, or function within the weapon (all cartridge types).
b) Metal parts separation.
c) A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

Note 7: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

Aircraft Ammunition - Pyro					
Item[1,4,5]	**Weapon Interface[6]**	**Function and Casualty/Safety Order size greater than 100 cartridges Sample Size[2]-Accept-Reject & Retest (cumulative)**			**Function and Casualty/Safety Order size 1-100 cartridges[3] Sample Size[2]-Accept-Reject & Retest (cumulative)**
		Ambient	Hot	Cold	Cold
PP3	-	15-0-1 Retest 29-1-2 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)
PP9					
PPL	YaKB				
EKSR-46 Flare	Mi-17/35	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)
PPI-26 Flare	ASO-2/ ASO-2V				

Note 1: Any Pyro cartridges not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 15 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: Due to the intended use of these pyro devices testing shall be conducted at cold for order quantities of less than 100.

Note 4: All cartridges shall function per the manufacturer's specification. A function failure shall be defined as:

a) A cartridge that will not load into the firing device or weapon.
b) A misfire (failure of a projectile to discharge as intended) or failure to ignite.
c) A cartridge that does not meet manufacturer's minimum requirements (example: light intensity, pressure, velocity, duration…).

Note 5: Critical safety failures are defined as:

a) Detonation or function within the weapon.
b) A cartridge/item that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

Note 6: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

Tank and Gun Ammunition					
Item[1,4,5,6]	**Weapon[7]**	**Function and Casualty/Safety Order size greater than 250 cartridges Sample Size[2]-Accept-Reject & Retest (cumulative)**			**Function and Casualty/Safety Order size 100 – 250 cartridges[3] Sample Size[2]-Accept-Reject & Retest (cumulative)**
		Ambient	Hot	Cold	Ambient Only
100mm APFSDS, HEAT & HE-Frag	T-55 Tank	15-0-1 Retest 29-1-2 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)
115mm APFSDS, HEAT & HE Frag	T-62 Tank				
125mm APFSDS, HEAT & HE-Frag	T-64/T-72 Tank				
73mm HEAT & HE-Frag	2A28 Grom				
Notes on next page					

Note 1: Any tank, and gun ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 15 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 4: Not less than 80% of cartridges containing tracers (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's requirements. All tracers within the lot shall be of the same color. If the lot fails to meet the Tracer requirements, a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes.

Note 5: A function failure shall be defined as:

- a) A cartridge that will not load into the weapon.
- b) A misfire - failure of a projectile to discharge as intended.
- c) A short round: A cartridge that has half of the muzzle velocity of the others within its group and/or impacts the ground at a 50% less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
- d) A cartridge that does not completely detonate upon initial impact with their target (HE & HEAT cartridges only). Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test.
- e) A cartridge that does not have complete penetration of armor per manufacturer's requirements (AP & HEAT cartridges only).
- f) A cartridge that does not perform its primary function.

Note 6: Critical safety failures are defined as:

- a) Detonation, Deflagration, or function within the weapon (all cartridge types).
- b) Detonation of HE or HEAT before impact with the ground or target.
- c) Metal parts separation.
- d) A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

Note 7: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

Missiles					
Item[1,4,5,6]	**Weapon[7]**	**Function and Casualty/Safety Order size greater than 250 cartridges Sample Size[2]-Accept-Reject & Retest (cumulative)**			**Function and Casualty/Safety Order size 100 – 250 cartridges[3] Sample Size[2]-Accept-Reject & Retest (cumulative)**
		Ambient	Hot	Cold	Ambient Only
AT-3 ATGM HE, (Sagger)	BMP-1, launcher 9P111, BRDM Vehicle	15-0-1 Retest 29-1-2 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)
AT-5 ATGM HE (Spandrel)	BMP-2/BRDM-2				
AT-7 ATGM HE, (Saxhorn)	9P151				

Note 1: Any ATGM ammunition not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 15 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 3: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 4: Not less than 80% of cartridges containing tracers (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's requirements. All tracers within the lot shall be of the same color. If the lot fails to meet the Tracer requirements, a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes.

Note 5: A function failure shall be defined as:

a) A cartridge that will not load into the weapon or launcher.
b) A misfire - failure of a projectile to discharge as intended.
c) A short round: A cartridge that has half of the muzzle velocity of the others within its group and/or impacts the ground at a 50% less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
d) A cartridge that does not completely detonate upon initial impact with their target (HE & HEAT cartridges only). Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test.
e) A cartridge that does not have complete penetration of armor per manufacturer's requirements (AP & HEAT cartridges only).
f) A cartridge that does not perform its primary function.

Note 6: Critical safety failures are defined as:

a) Detonation of HE within the weapon or launcher, or before impact with the ground or target.
b) Metal parts separation.
c) A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

Note 7: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

Hand Grenades				
Item / Requirement[1,6,8]	**Function and Casualty/Safety Order size greater than 250 Sample Size[2]-Accept-Reject & Retest (cumulative)**			**Function and Casualty/Safety Order size 100 – 250 cartridges[7] Sample Size[2]-Accept-Reject & Retest (cumulative)**
	Ambient	Hot	Cold	Ambient Only
RKG-3 / Armor penetration >=150mm[5] Grenade, Rubber Buckshot[3] Grenade, Flash & Bang[4] Grenade, HE, HE Frag. or Thermobaric[3]	32-0-1 Retest 60-1-2 (Cumulative)	32-0-1 Retest 60-1-2 (Cumulative)	32-0-1 Retest 60-1-2 (Cumulative)	32-0-1 Retest 60-1-2 (Cumulative)
Grenade, Smoke Grenade, Tear Gas	15-0-1 Retest 29-1-2 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-0-1 Retest 29-1-2 (Cumulative)

Note 1: Hand grenades not listed in this table are still required to meet sample size-accept-reject criteria listed in this table.

Note 2: These samples sizes shall be used to show performance and function and casualty at ambient temperature. The rounds shall be tested for function and safety at the operating temperature extremes. A single critical safety failure will result in lot rejection and there shall be no retest.

Note 3: These grenades must be tested at hot/cold/ambient for each lot (skip lot testing does not apply). A critical safety failure is grenade that exhibits a delay of less than 3 seconds. The lot shall be rejected; there shall be no retest.

Note 4: These grenades must be tested at hot/cold/ambient for each lot (skip lot testing does not apply). A critical safety failure is grenade that exhibits a delay of less than 1 second. The lot shall be rejected; there shall be no retest.

Note 5: These grenades must be tested at hot/cold/ambient for each lot (skip lot testing does not apply). A critical safety failure is grenade that detonates prior to impact. The lot shall be rejected; there shall be no retest.

Note 6: All applicable grenades shall completely detonate. Fuze function only shall be scored as a function failure.

Note 7: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 8: Additional critical safety failures which apply to all hand grenades are defined as:

a) Metal parts separation.
b) A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

ANNEX A - Weapon Interface and Sample Size Requirements (continued)

Links, Linked Belts, and Linking Machines			
Item[1]	**Caliber**	**Weapon Interface**	**Function and Casualty/Safety Sample Size-Accept-Reject & Retest (cumulative)**
			Ambient Only
Links	20mm and larger	GSh-23, ZU-23, 2A42, Gsh-30	50-1-2 Retest 100-2-3 (Cumulative)[2,3]
Linked Belts	14.5mm and smaller	RPD, KPV, PKM, PKT, YaKB, DshKM	200-1-2 Retest 400-2-3 (Cumulative)[2,3]
Linking Machines	All calibers of links and belts	-	5-0-1 Retest 10-1-2 (Cumulative)[4,5]

Note 1: Any size/caliber link, belt, or linking machine not listed in this table is still required to meet sample size-accept-reject criteria listed in this table.

Note 2: For orders less than 10,000 individual links or 200 belts, function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 3: These samples sizes shall be used to show performance and function and casualty at ambient temperature. The sample size represents the total number of cartridges fired (not belts). The number of belts used for function and casualty will be determined by the number of rounds they hold (example: if 12.7x108mm 50 round belts are being tested, a total of 4 belts shall be fired to meet a requirement of 200 cartridges). For individual links, the contractor can determine how many cartridges will be linked together to form a belt, however, a belt shall contain not less than 10 cartridges. Linked belts shall be linked with live ammunition and function tested in the weapons system the ammunition and links are designed to function.

a) Any type of weapon stoppage that cannot be attributed to the ammunition or weapon shall be counted as a single link/belt failure.
b) Links/belts that damage the weapon or cause weapon stoppages which cannot be cleared without use of special tools, rendering it inoperable, are critical failures. The lot shall be rejected. No retest allowed.

Note 4: For orders of 25 or greater linking machines, a minimum of 5 linking machines shall be tested for failure free operation. For orders under 25 linking machines, a minimum of 3 linking machines, shall be tested with an Accept-Reject criteria of 3-0-1 and Retest of 6-1-2 (cumulative). If the order quantity is less than 3 then all Linking machines shall be tested and Accept-Reject criteria shall be X-0-1. The USG reserves the right to increase these requirements in their proposals.

Note 5: A minimum of 100 cartridges shall be cycled through the linking machine regardless of caliber and belt length. Ammunition processed through the linking machine shall be firmly positioned in the linked belt and be capable of functioning in the specific weapon. The cartridges shall be seated such that there is no gap between the mouth of the casing and free hanging portion of the link, or the link seats within the rim of the cartridge.

a) Any stoppages or jams due to design or manufacturing defect of the linking machine shall be counted as a single failure. Any damage to the links or ammunition caused by the linking machine shall be counted as a single failure.
b) Linking machines that damage links/belts/ammunition rendering them unable to be fired in a weapon are critical failures. Linking machines that cause ammunition to fire, function, activate, or become unsafe during loading of the links are critical failures. If a critical occurs the lot shall be rejected. No retest allowed.

ANNEX B – Visual Inspection Requirements for Ammunition

Critical: 24-0-1[2,3]	**Major: 24-1-2[2,3]** **Reinspection: 72-3-4 (cumulative)[1,3]**	**Minor: 24-3-4[2,3]** **Reinspection: 72-9-10 (cumulative)[1,3]**
Missing, cracked, loose or bent parts	Cases with moving cartridges; loose pack	Discolored, oily, dirty, or smeared
Gross Corrosion / Pitting	Incorrect quantity of cartridges in packaging (light boxes shall have their corners painted) each cartridge missing shall constitute one failure	Minor surface corrosion
Spillage/leakage of energetic materials	Wire Security Seal missing	Dents, chips or burrs
Incorrect Marking of cartridge type: HE cartridge marked as Practice, Smoke or Illumination	Mixed ammunition types: Training cartridge mixed with Combat cartridge lot	Scratches on cartridge
Gross deformation	Missing, illegible or incorrect marking	Excessive grease on cartridge
Splits in the case around primer area, case groove or case rim (14.5, 23 and 30mm cartridges only)	Incorrect Marking of cartridge type: Practice, Smoke or Illumination cartridges marked incorrectly	Irregular disposition of cartridges in case
Mixed ammunition types: Combat cartridge mixed with training cartridge lot	Projectile weight is out of specified range	Paint damaged, wet or incomplete
Bourrelet diameter too large	Primer missing, inverted, double, cocked, or visually above flush	

Note 1: If the number of allowable defects is exceeded a double sample quantity of 48 cartridges shall be re-inspected. The accept/reject criteria is based on the cumulative number of defects for all three samples.

Note 2: If a critical is found or the number of allowable majors or minors, after reinspection, is exceeded the lot fails.

Note 3: The inspection and reinspection quantity listed is selected during LAT. DCMA will utilize their Zero-Based Acceptance Sample Plans shown in Appendix during GSI.

ANNEX B – Visual Inspection Requirements for Links, Linked Belts and Linking Machines

Belts and Linking Machines **Critical: 5-0-1**	**Belts and Linking Machines** **Major: 5-1-2** **Reinspection: 15-3-4 (cumulative)**	**Belts and Linking Machines** **Minor: 5-3-4** **Reinspection: 15-9-10 (cumulative)**
Links **Critical: 24-0-1**	**Links** **Major: 24-1-2** **Reinspection: 72-3-4 (cumulative)**	**Links** **Minor: 24-3-4** **Reinspection: 72-9-10 (cumulative)**
Gross corrosion or pitting	Minor surface corrosion	Discolored, excessive oil, dirty, or smeared
Gross deformation	Missing wire security seal	Dents, folds, or wrinkles
Missing, cracked, or loose parts	Missing, illegible, or incomplete markings	Scaly, scratched, or chipped metal
	Incorrect quantity of links in belt (belts only)	Sharp edges
	Incorrect quantity of items in packaging	Surface coating wet, damaged, or incomplete
	Mixed or incorrect caliber type	

Note 1: A minimum of 5 belts/linking machines shall be visually inspected per lot. For quantities less than 5 the entire lot must be inspected.

Note 2: A minimum of 24 links shall be visually inspected per lot.

Note 3: If the number of allowable defects is exceeded a double sample quantity shall be re-inspected. The accept/reject criteria is based on the cumulative number of defects for all three samples.

Note 4: If a critical is found or the number of allowable majors or minors, after reinspection, is exceeded the lot fails.

Note 5: The inspection and reinspection quantity listed, is selected during LAT. DCMA will utilize their Zero-Based Acceptance Sample Plans shown in Appendix during GSI.

APPENDIX

1. DEFINITIONS

1.1 PdL SAWS Special Ammunition and Weapon Systems. PdL SAWS Special Ammunition and Weapon Systems are those munitions that have not been safety tested and type classified for Army use, and cannot be procured through the Army supply. Munitions and explosives that are not managed by National Inventory Control Points, have not been safety tested nor type classified for Army use, do not have a national stock number (NSN) and cannot be procured or requisitioned through the Army or other Department of Defense supply system.

1.2 Technical data. Technical data, otherwise known as the Technical Data Package (TDP), is the product specific technical drawings and Quality Assurance (QA) requirements to which ammunition and associated packaging are produced and accepted for each applicable Contract Line Item Number (CLIN). For the purpose of this specification, the TDP, as a minimum, must contain the following: Government Contract Number and Delivery Order; top assembly, fuze and key component drawings with revision number, revision date, document control signature, key interface dimensions, and a list of component assemblies with drawing numbers and revision dates; packaging and marking drawings with revision number, revision date, document control signature and markings; and product specification with final assembly, fuze and key component in-process acceptance test methods with sample size, and accept/reject criteria, and retest criteria. Within the TDP there shall also be a clear cut-away and/or exploded drawing(s) to show all of the inner parts and workings of the fuze. Additionally, there shall be paragraphs which describe the fuze arming sequence and events leading to function. A copy of the manufacturer's firing tables for ammunition shall be included with the TDP.

1.3 Non-Energetic Items. Items such as links, belts, and linking machines which are associated with ammunition and weapon systems but do not fire a projectile or contain pyrotechnics or other energetic materials.

1.4 Degradation. Ammunition with gross nonconformance to identified technical requirements, corrosion, cracks, deformation, and spillage

1.5 Deterioration. Packaging ripped, broken, perforated, water damaged, or crushed.

1.6 Skip-Lot Temperature Testing. Except for the hand grenades identified in ANNEX A which require temperature testing on all lots, testing at the operational temperature limits shall be conducted on the first lot of every delivery order and every 5th lot thereafter (1, 5, 10, 15...). The ammunition must pass the requirements defined in ANNEX A. If a lot is rejected for any reason, the following two lots must pass temperature testing before resuming the skip lot procedures.

APPENDIX (continued)

	Acceptable Quality Level (AQL)															
LOT SIZE	**0.01**	**0.015**	**0.025**	**0.04**	**0.065**	**0.10**	**0.15**	**0.25**	**0.40**	**0.65**	**1.0**	**1.5**	**2.5**	**4.0**	**6.5**	**10.0**
1-8	A	A	A	A	A	A	A	A	A	A	A	A	5	3	3	3
9-15	A	A	A	A	A	A	A	A	A	A	13	8	5	3	3	3
16-25	A	A	A	A	A	A	A	A	A	20	13	8	5	3	3	3
26-50	A	A	A	A	A	A	A	A	32	20	13	8	7	7	5	3
51-90	A	A	A	A	A	A	80	50	32	20	13	13	11	8	5	4
91-150	A	A	A	A	A	125	80	50	32	20	19	19	11	9	6	5
151-280	A	A	A	A	200	125	80	50	32	29	29	19	13	10	7	6
281-500	A	A	A	315	200	125	80	50	48	47	29	21	16	11	9	7
501-1200	A	800	500	315	200	125	80	75	73	47	34	27	19	15	11	8
1201-3200	1250	800	500	315	200	125	120	116	73	53	42	35	23	18	13	9
3201-10,000	1250	800	500	315	200	192	189	116	86	68	50	38	29	22	15	9
10,001-35,000	1250	800	500	315	300	294	189	135	108	77	60	46	35	29	15	9
35,001-150,000	1250	800	500	490	476	294	218	170	123	96	74	56	40	29	15	9
150,001-500,000	1250	800	750	715	476	345	270	200	156	119	90	64	40	29	15	9
500,001 & Over	1250	1200	1112	715	556	435	303	244	189	143	102	64	40	29	15	9

Note: An "A" in place of a number indicates that all units in a lot must be inspected